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(54) **FUEL INJECTOR OF AN INTERNAL COMBUSTION ENGINE AND COMBUSTION ENGINE**

(56) **References Cited**

U.S. PATENT DOCUMENTS

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2,530,128 A * 11/1950 Mashinter F02M 59/14
417/520
2007/0210189 A1 * 9/2007 Schurz F02M 61/20
239/533.9

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FOREIGN PATENT DOCUMENTS

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DE	10054182	5/2002	
DE	10344880	4/2005	
DE	102007018626	10/2008	
DE	102007018626 A1 *	10/2008	 F02M 51/005
DE	102008044153 A1 *	6/2010	 F02M 51/0603
DE	102011117534 A1 *	5/2013	 F02M 55/04
DE	102020208273	1/2022	
DE	102020210145 A1 *	2/2022	
DE	102022206220	12/2023	
EP	1445473 A1 *	8/2004	 F02M 51/0603
EP	1519036	3/2005	

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OTHER PUBLICATIONS

DE 102007018626 A1—English Translation (Year: 2008).*

(Continued)

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63/0012

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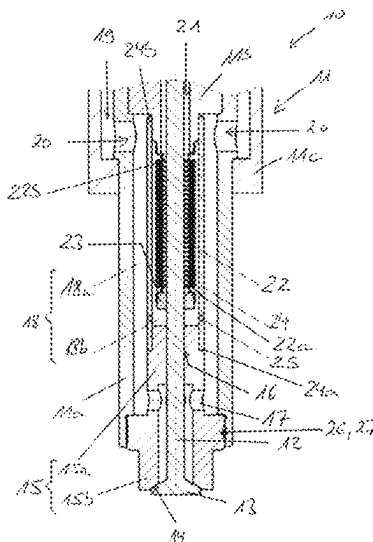
See application file for complete search history.

(57)

ABSTRACT

A fuel injector for an internal combustion engine, which is configured for feeding fuel to a cylinder, having a housing, via which the fuel injector can be mounted to the internal combustion engine, a valve body engaging with a valve stem interacting with a valve seat, which can be moved relative to the valve seat for opening a fuel flow, having a bellows sealing element, which with a first end engages with the valve stem and with a second end with the housing, a sleeve body, which at least partly decouples the bellows sealing element from a fuel flow through the fuel injector and from fuel pressure pulsations.

10 Claims, 1 Drawing Sheet



(56)

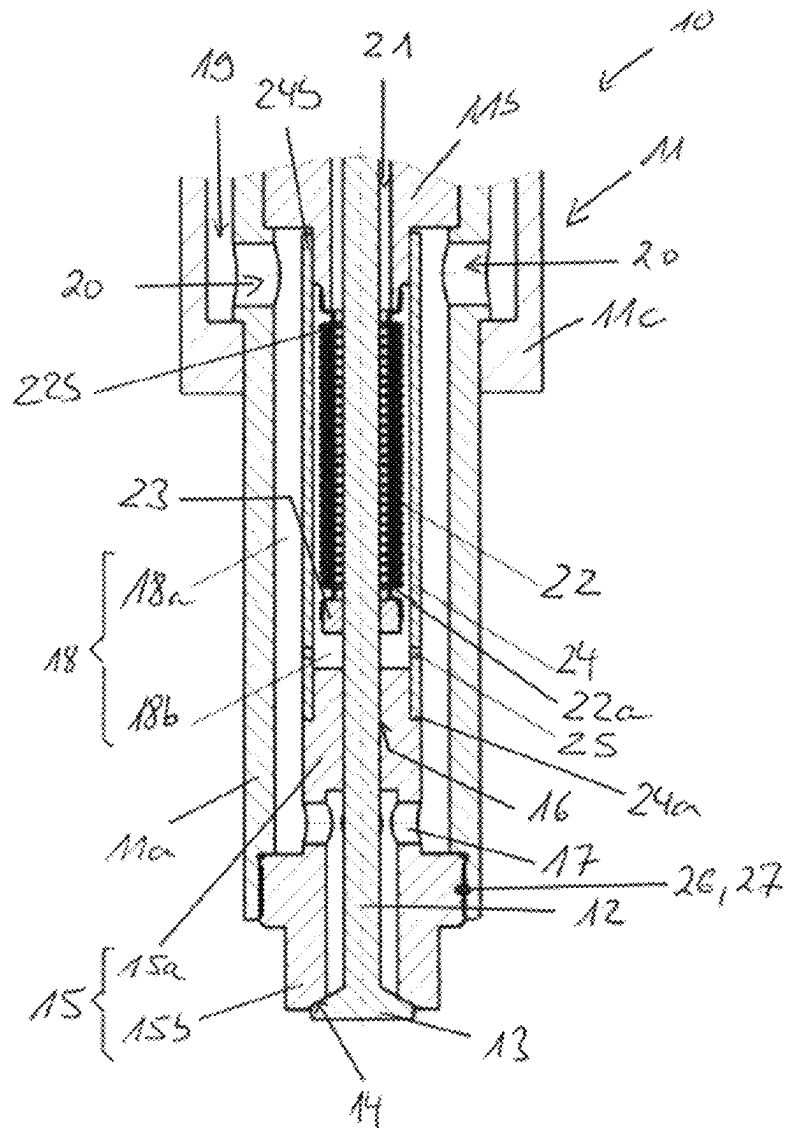
References Cited

OTHER PUBLICATIONS

Office Action of corresponding German Patent Application No. 10
2023 111 573.1.

European Search Report of corresponding European Patent Appli-
cation No. 24161632.5, dated Sep. 6, 2024.

* cited by examiner



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FUEL INJECTOR OF AN INTERNAL COMBUSTION ENGINE AND COMBUSTION ENGINE

BACKGROUND OF THE INVENTION

1. Field of the Invention

The disclosure relates to a fuel injector of an internal combustion engine and an internal combustion engine with such a fuel injector.

The disclosure relates in particular to so-called large engines or large internal combustion engines, the cylinders of which have piston diameters of at least 140 mm, in particular of at least 175 mm. Such large internal combustion engines are for example marine engines.

2. Description of Related Art

From practice, gas engines, liquid fuel engines or large internal combustion engines designed as dual-fuel engines are known. In liquid fuel internal combustion engines, a liquid fuel, such as for example a diesel fuel, is combusted. In internal combustion engines, a gaseous fuel, such as for example natural gas, is combusted. In dual-fuel engines, a liquid fuel can be combusted in a first operating mode and a gaseous fuel in a second operating mode.

In order to feed the fuel to cylinders of the internal combustion engine, internal combustion engines comprise fuel injectors. By way of a fuel injector, the fuel can be directly introduced into a combustion space or a combustion chamber of a cylinder or into a pre-chamber of the cylinder interacting with the combustion space or combustion chamber or into an intake pipe leading to the cylinder.

Fuel injectors have housing, via which the respective fuel injector can be mounted to an internal combustion engine. Further, fuel injectors have a valve body interacting with a valve stem, which valve body interacts with a valve seat. Dependent on the relative position between valve body and valve seat, a fuel flow is either opened or blocked.

In fuel injectors, a basic distinction is made between fuel injectors having a valve body opening towards the outside or a valve body opening towards the inside. With a fuel injector having a valve body opening towards the outside, the valve body, for opening a fuel flow, can be moved in a direction away from the housing or based on the housing, in the direction towards the outside. With a fuel injector having a valve body opening towards the inside, the valve body, by contrast, can be moved for opening the fuel flow, based on the housing, in the direction towards the inside into the housing. The invention can be employed with a fuel injector having a valve body opening towards the outside for opening the fuel flow, or even with a fuel injector having a valve body opening to the inside for opening the fuel flow.

SUMMARY OF THE INVENTION

There is a need for a fuel injector of an internal combustion engine having a longer lifespan.

Starting out from this, the present invention is based on the object of creating a new type of fuel injector of an internal combustion engine and an internal combustion engine having such a fuel injector.

The fuel injector comprises a housing via which the fuel injector can be mounted to the internal combustion engine. The fuel injector comprises a valve body interacting with a valve stem and interacting with a valve seat, which valve

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body, for opening a fuel flow, can be moved relative to the valve seat. The fuel injector comprises a bellows sealing element, which with a first end engages with the valve stem and with a second end with the housing. Further, the fuel injector comprises a sleeve body which at least partially decouples the bellows sealing element from a fuel flow through the fuel injector and from fuel pressure pulsations. By way of the sleeve body, the bellows sealing element can be protected from loads in particular through fuel pressure pulsations. The sleeve body decouples the bellows sealing element at least partially from the fuel flow. Pressure fluctuations or pressure pulsations within the fuel flow do not have a direct effect on the bellows sealing element. Thus, the fatigue strength of the bellows sealing element and thus the lifespan of the fuel injector can be increased.

Preferentially, the sleeve body divides a fuel space within the housing into first sub-space and a second sub-space, wherein the first sub-space can be flowed through by the fuel flow, and wherein the bellows sealing element is arranged in the second sub-space. Thus, fatigue strength of the bellow sealing element and thus the lifespan of the fuel injector can be particularly advantageously increased.

Preferentially, the first sub-space of the fuel space and the second sub-space of the fuel space are connected via at least one opening or bore or choke in the sleeve body. Thus it is possible to offset a volume change of fuel in the second sub-space caused by a stroke movement of the bellows sealing element, in which the bellows sealing element is arranged, in that fuel can flow via the at least one opening or bore or choke.

Preferentially, the first sub-space of the fuel space is formed between the housing and the sleeve body radially outside adjoining the sleeve body, wherein the second sub-space of the fuel space is formed radially inside adjoining the sleeve body, and wherein the sleeve body surrounds the bellows sealing element radially outside in such a manner that the bellows sealing element is arranged within the sleeve body. This design is particularly preferred in order to increase the fatigue strength of the bellows sealing element and thus the lifespan of the fuel injector.

Preferentially, the fuel injector comprises a guide and seat body with a guide portion and a seat portion, which is mounted to the housing or an integral part of the housing, wherein the guide portion of the guide and seat body guides the valve stem and the seat portion of the guide and seat portion provides the valve seat, and wherein the sleeve body, with a first end engages with the guide and seat body and with a second end with the housing. This embodiment is structurally simple.

Preferentially, the sleeve body is sealing connected in each case at its first end to the guide and seat body and at its second end to the housing. A coupling of the two sub-spaces of the fuel space takes place merely via the at least one opening or bore or choke. This is preferred in order to increase the fatigue strength of the bellows sealing element and thus the lifespan of the fuel injector.

Preferentially, the bellows sealing element is indirectly connected at its first end to the valve stem via a ring or shoulder interacting with the valve stem. This design is structurally particularly simple.

BRIEF DESCRIPTION OF THE DRAWINGS

Preferred further developments of the invention are obtained from the following description. Exemplary

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embodiments of the invention are explained in more detail by way of the drawing without being restricted to this. There it shows:

The FIGURE: is a schematic cross-section through a fuel injector of an internal combustion engine.

DETAILED DESCRIPTION OF THE PRESENTLY PREFERRED EMBODIMENTS

The FIGURE shows a cross-section through an exemplary embodiment of a fuel injector **10** for an internal combustion engine according to the invention. The fuel injector **10** can be both a fuel injector for liquid fuel and also a fuel injector for gaseous fuel. The fuel injector **10** is configured in order to feed fuel to a cylinder of the internal combustion engine, namely either directly into a combustion chamber of the cylinder or into a pre-chamber of the cylinder interacting with the combustion chamber or into a pipe of the internal combustion engine leading to the combustion chamber of the cylinder, which also carries combustion air to the cylinder.

The fuel injector **10** has a housing **11**. Via the housing **11**, the fuel injector **10** can be mounted to the internal combustion engine, for example in the region of a cylinder head or a cylinder head cover. Furthermore, the fuel injector **10** has a valve stem **12**, which carries a valve body **13**. The valve stem **12** can be directly actuated magnetically. The valve body **13** interacts with a valve seat **14**. In particular when the valve body **13** lies against the valve seat **14**, the fuel injector **10** does not allow any fuel flow. When by contrast the valve body **13** is lifted off the valve seat **14**, the fuel injector **10** opens a fuel flow.

Furthermore, the fuel injector **10** has a guide and seat body **15** with a guide portion **15a** for the valve stem **12** and a seat portion **15b** providing the valve seat **14**.

In the shown exemplary embodiment, this guide and seat body **15** is embodied relative to the housing **11** as separate, integral or monolithic assembly and mounted to the housing **11**. According to the FIGURE, the guide and seat body **15** and the housing **11** are screwed to the guide and seat body **15**, in particular to the seat portion **15b** of the same via interacting threads, namely an internal thread **26** on the housing **11** and an external thread **27** on the guide and seat body **15**.

As already explained, the guide portion **15a** of the guide and seat body **15** guides the valve stem **12**, wherein for this purpose the valve stem **12** penetrates a recess **16** in the guide portion **15a**.

The seat portion **15b** of the guide and seat body **15** provides the valve seat **14** for the valve body **13**.

The guide portion **15a** of the guide and seat body **15** can be flowed about by fuel. By contrast, the seat portion **15b** of the guide and seat body **15** can be flowed through by the fuel. When the valve body **13** is lifted off the valve seat **14** and the fuel injector **10** opens a fuel flow, the fuel flows about the guide portion **15a** and flows through the seat portion **15b** of the guide and seat body **15**.

By way of expansions or bores **17**, the fuel, with opened fuel injector **10**, emanating from the guide portion **15a**, can overflow into the seat portion **15b**.

It is pointed out here that the guide and seat body **15** can also be an integral part of the housing **11**. The guide and seat body **15** can also be divided into multiple bodies, namely into a guide body, which guides the valve stem **12**, and into a seat body that is separate relative to the guide body, which provides the valve seat.

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Multiple parts or portions **11a**, **11b** and **c** of the housing **11** are shown. In the shown exemplary embodiment, the guide and seat body **15** engages with a first end of the housing part or housing portion **11a**. Accordingly, the internal thread **26** which interacts with the external thread **27** of the guide and seat body **15** is formed on this end of the housing part or housing portion **11**, in order to fasten the guide and seat body **15** to the housing part or housing portion **11a** by screw connection. The housing portion **11a** delimits radially outside a fuel space **18** of the fuel injector, wherein the fuel, emanating from an annular gap **19** formed between the housing part or housing portion **11a** and the housing part or housing portion **11c** can overflow via openings or bores **20**. The housing part or the housing portion **11b** is received in the housing part or housing portion **11a**. The valve stem **12**, which is guided in the guide portion **15a** of the guide and seat body **15**, penetrates the fuel space **18**, which is delimited by the housing part or housing portion **11a** radially outside. Further, the valve stem **12** penetrates the housing part or housing portion **11b**.

In order to prevent that fuel, emanating from the fuel space **18**, flows via a gap **21** formed between the valve stem **12** and the housing part or the housing portion **11b**, a bellows sealing element **22** is arranged within the fuel space **18**. The bellows sealing element **22** engages with a first end **22a** with the valve stem **12** and with an opposite end **22b** with the housing **11**, namely with the housing part or housing portion **11b**. The connection of the first end **22a** of the bellows sealing element **22** to the valve stem **12** and the connection of the second end **22b** of the bellows sealing element **22** to the housing **11**, namely to the housing part or housing portion **11b**, is fluid-tight in each case. Because of the bellows structure of the bellows sealing element **22**, the bellows sealing element **22** can offset or compensate for the relative movement between valve stem **12** and housing **11** upon a stroke movement of the valve stem **12**.

The bellows sealing element **22**, in the shown exemplary embodiment, engages with its first end **22a** directly with the valve stem **12**, namely with a ring or shoulder **23**, which is connected to the valve stem **12** firmly and in a fluid-tight manner. The opposite second end **22** of the bellows sealing element **22** engages directly with the housing **11**, namely with the housing part or housing portion **11b**, in a fluid-tight manner. The ring or shoulder **23** can also be an integral part of the valve stem **12**.

It is not only the bellows sealing element **22** that is arranged in the fuel space **18** but in addition a sleeve body **24**. In the shown exemplary embodiment, the sleeve body **24** with a first end **24a** engages with the guide and support body **15**, namely with the guide portion **15a** of the same, and with a second, opposite end **24b**, with the housing **11**, namely with the housing part or housing portion **11b**.

The sleeve body **24** decouples the bellows sealing element **22** from the fuel flow through the fuel injector **10** and from pressure pulsations of the fuel at least partially. Thus, it can be ensured that pressure fluctuations in the fuel space **18** do not have a direct effect on the bellows sealing element **22**. Accordingly, the bellows sealing element **22** can be protected in particular from pressure peaks, as a result of which the fatigue strength of the bellows sealing element and thus the lifespan of the fuel injector **10** are increased.

The sleeve body **24** arranged in the fuel space **18** divides the fuel space **18** into a first sub-space **18a** and a second sub-space **18b**. The first sub-space **18a** can be flowed through by the fuel flow, while the bellows sealing element **22** is arranged in the second sub-space **18b**. The first sub-space **18a** is formed between the housing, namely the

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housing part or housing portion 11a, and the sleeve body 24, the second sub-space 18b is arranged radially within the sleeve body 24. The first sub-space 18a of the fuel space 18 accordingly adjoins the sleeve body 24 radially outside, the second sub-space 18b the sleeve body 24 radially inside. 5

In the shown exemplary embodiment, the sleeve body 24 with its first end 24a is connected to the guide and seat body 15 in a sealing manner, on the opposite second end 24b, to the housing in a sealing manner, namely the housing part or housing portion 11b. The sleeve 24 is a rigid sleeve which does not have to offset a relative movement between the housing part or housing portion 11b and the guide and seat body 15, since such a relative movement does not occur. 10

The first sub-space 18a of the fuel space 18 is connected to the second sub-space 18b of the fuel space 18, preferentially via at least one opening or bore or choke 25 in the sleeve body 24. Thus, a change in volume of fuel in the sub-space 18b caused by a stroke movement of the valve stem 12 and thus of the bellows sealing element 22 can be offset. 20

The bellows sealing element 22 is preferentially an element of a metallic material. The sleeve body 24 can be produced from a metallic material or from a plastic.

The sleeve body 24 avoids or reduces the effect of the fuel pressure on the bellows sealing element 22. Thus, the fatigue strength of the bellow sealing element 22 is increased. The lifespan of the fuel injector 10 can be increased. Further it is possible to expand operating parameters, in particular utilise higher fuel pressures. Higher fuel pressures can be absorbed by the sleeve body 24. A direct fuel inflow onto the bellows sealing element 22 is prevented by the sleeve body 24. The same is thus shielded against the fuel flow as a result of which the fatigue strength of the same is increased. During the mounting, the bellow sealing element 22 can be protected by the sleeve body 24. 35

Thus, while there have shown and described and pointed out fundamental novel features of the invention as applied to a preferred embodiment thereof, it will be understood that various omissions and substitutions and changes in the form and details of the devices illustrated, and in their operation, may be made by those skilled in the art without departing from the spirit of the invention. For example, it is expressly intended that all combinations of those elements and/or method steps which perform substantially the same function in substantially the same way to achieve the same results are within the scope of the invention. Moreover, it should be recognized that structures and/or elements and/or method steps shown and/or described in connection with any disclosed form or embodiment of the invention may be incorporated in any other disclosed or described or suggested form or embodiment as a general matter of design choice. It is the intention, therefore, to be limited only as indicated by the scope of the claims appended hereto. 50

The invention claimed is:

1. A fuel injector for an internal combustion engine, and configured to feed fuel to a cylinder of the internal combustion engine, comprising:

a housing, via which the fuel injector can be mounted to the internal combustion engine;

a valve stem;

a guide and seat body comprising a valve seat;

a valve body that engages with the valve stem and which interacts with the valve seat, and which for opening a fuel flow is moveable relative to the valve seat;

a bellows seal, which with a first end that engages with the valve stem and a second end that engages with the housing; and 65

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a sleeve body which with a first end that engages with the guide and seat body and a second end that engages with the housing and which at least partially decouples the bellows seal from a fuel flow through the fuel injector and from fuel pressure pulsations,

wherein

the sleeve body divides a fuel space within the housing into a first sub-space and a second sub-space, the fuel flow to the valve seat is entirely through the first sub-space, and

the bellows seal is arranged in the second sub-space.

2. The fuel injector according to claim 1, wherein the first sub-space of the fuel space and the second sub-space of the fuel space are connected via at least one opening or bore or choke in the sleeve body.

3. The fuel injector according to claim 1, wherein the first sub-space is formed between the housing and the sleeve body adjoining the sleeve body radially outside, and

the second sub-space is formed adjoining the sleeve body radially inside.

4. The fuel injector according to claim 1, wherein the sleeve body surrounds the bellows seal radially outside in such a manner that the bellows seal is arranged within the sleeve body.

5. The fuel injector according to claim 1, wherein the guide and seat body comprises a guide portion and a seat portion, which is mounted to the housing or is an integral part of the housing, wherein a guide portion of the guide and seat body guides the valve stem and a seat portion of the guide and seat body provides the valve seat.

6. The fuel injector according to claim 5, wherein the sleeve body at its first end is sealingly connected to the guide and seat body and sealingly connected at its second end to the housing.

7. The fuel injector according to claim 1, wherein the bellows seal is sealingly connected at its first end to the valve stem and sealingly connected at its second end to the housing.

8. The fuel injector according to claim 1, wherein the bellows seal is indirectly connected at its first end to the valve stem via a ring or a shoulder interacting with the valve stem.

9. The fuel injector according to claim 1, wherein the bellows seal is a metallic material, and wherein the sleeve body is a metallic material or a plastic.

10. An internal combustion engine, comprising: cylinders; and

a fuel injector for each cylinder comprising:

a housing, via which the fuel injector can be mounted to the internal combustion engine;

a valve stem;

a guide and seat body comprising a valve seat;

a valve body that engages with the valve stem and which interacts with the valve seat, and which for opening a fuel flow is moveable relative to the valve seat;

a bellows seal, which with a first end that engages with the valve stem and a second end that engages with the housing; and

a sleeve body which with a first end that engages with the guide and seat body and a second end that engages with the housing and which at least partially decouples the bellows seal from a fuel flow through the fuel injector and from fuel pressure pulsations,

wherein

the sleeve body divides a fuel space within the housing
into a first sub-space and a second sub-space,
the fuel flow to the valve seat to is entirely through the
first sub-space, and
the bellows seal is arranged in the second sub-space. 5

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